

SOFTWARE ENGINEERING
LAB 1
PROBLEM STATEMENT #59
COMMUNITY LOST AND FOUND MATCHING NETWORK

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REQUIREMENTS:

Problem Context

A community lost-and-found network that extracts visual and text item tags, performs geo-proximity matching between lost and found items, and verifies proof of ownership before revealing contact details.

Actors:

- finder/ owner
- Community admin

Functional requirements:

D	Type	Description	Priorit y	Acceptance Criteria	Rationale
FR -00 1	Function al	The system shall allow an owner to register a lost item by providing details such as category, description , color,	High	Pass: The lost-item report is successfully stored with all mandatory details. Fail: The system accepts a report with missing mandatory information.	Accurate lost-item informatio n is required for matching.

		date and location.			
FR-002	Functional	The system shall allow a finder to register a found item by providing details such as category, description, color, date and location.	High	Pass: The found-item report is successfully stored. Fail: The system accepts a report with missing mandatory information.	Found-item information is required to identify potential matches.
FR-003	Functional	The system shall match newly logged found items with active lost-item reports based on category, color tags and geographic	High	Pass: A match probability is calculated and compatible reports are identified. Fail: Incompatible item categories are matched.	Matching lost and found items is the core purpose of the system.

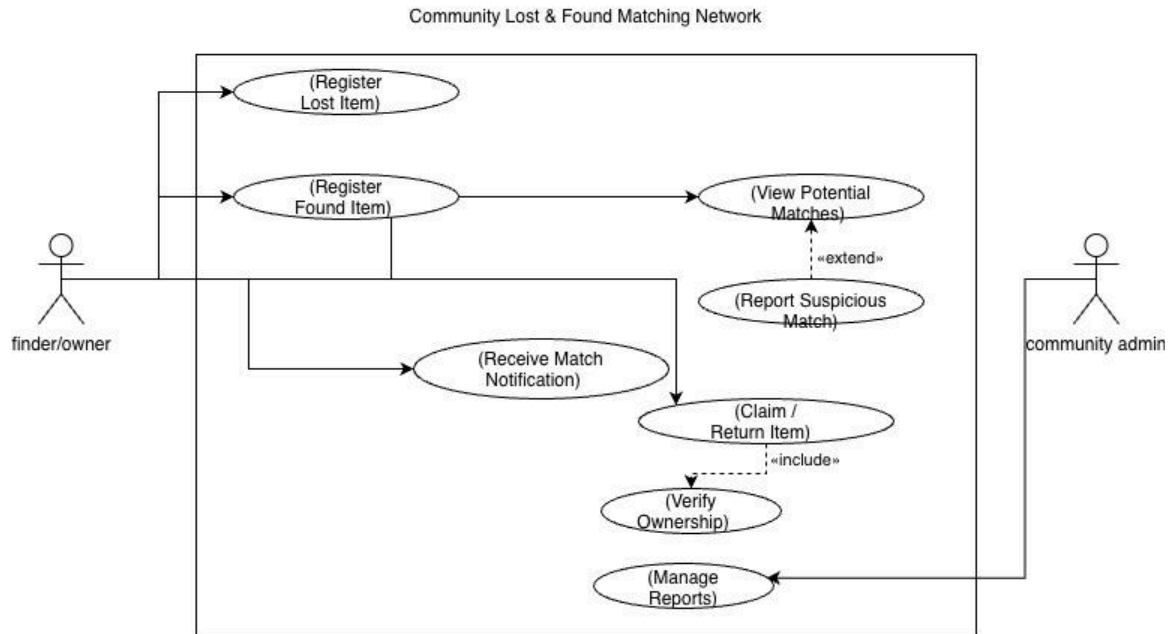
		al proximity.			
FR-004	Functional	The system shall notify potential owners when a found item has a sufficiently strong match with their lost-item report.	High	Pass: A notification is dispatched to the potential owner. Fail: No notification is generated for a valid match.	Users need to be informed when a potential match is identified.
FR-005	Functional	The system shall verify a claimant's answers to private security questions before revealing the finder's contact details.	High	Pass: Contact details are revealed only after successful verification. Fail: Contact details are revealed when verification fails.	Ownership verification helps prevent fraudulent claims and protects privacy.

Non functional requirements:

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NF R-0 01	Security	The system shall protect finder contact information and reveal it only after successful ownership verification.	High	Pass: Unauthorized users cannot access contact information. Fail: Contact information is displayed before successful verification.	Protects user privacy and reduces unauthorized access.

NF R-0 02	Performance	The system shall calculate potential matches and return results within the defined target response time under normal and peak usage.	Medium	Pass: Benchmarking confirms that the target response time is maintained under simulated peak load. Fail: Matching response time consistently exceeds the target.	Fast matching is necessary for the system to be practical for community users.
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UML DIAGRAM



USE CASE FLOW :

Use Case: Match Lost and Found Item

Primary Actor: Finder / Owner

Preconditions

1. The user has submitted a valid lost or found item report.
2. The report contains the required item details such as category, color, description, date, and location.
3. There are active lost or found reports available for comparison.

Postconditions

1. The system identifies potential matching items.
2. A match probability is calculated for compatible reports.
3. The relevant potential owner or finder is notified of the possible match.

Main Success Scenario

1. The Finder / Owner submits a lost or found item report.
2. The system validates the submitted item information.
3. The system retrieves active reports that may be compatible with the submitted item.
4. The system compares the item category and color tags.
5. The system compares the geographical proximity between the reported locations.
6. The system calculates the match probability.
7. The system identifies a potential match.
8. The system sends a notification to the relevant user.
9. The user views the potential match and can proceed with the ownership verification process.

Alternate Flow — No Suitable Match Found

1. The system compares the submitted item with active reports.
2. No report satisfies the matching criteria.
3. The system informs the user that no potential match was found.
4. The system keeps the report active so that it can be matched against future reports.

